TriLens: Per-Layer Logit-Lens Entropy for White-Box Hallucination Detection

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Abstract

When a language model hallucinates, the final answer is wrong, but the mistake is not necessarily invisible inside the model. Different internal pathways may remain uncertain, disagree in how quickly they sharpen, or commit to competing continuations before the output is produced. We introduce *TriLens*, a white-box detector that turns this intuition into a compact representation: at every layer, it reads the multi-head self-attention output, the feed-forward output, and the residual stream through the model's own logit lens, then records only the entropy of each readout. The resulting $3L$-dimensional trajectory summarizes how logit-lens concentration changes across depth and modules, without storing high-dimensional hidden states or sampling multiple generations. This signal yields a strong detector across three mid-sized instruction-tuned LLMs and four English QA benchmarks, and our analyses show that the three module-wise entropy trajectories provide complementary evidence. TriLens shows that hallucination detection in this controlled setting can benefit from layer-wise distributional concentration, not only the final-layer prediction. The code is available at https://tosakaucw.github.io/TriLens/.

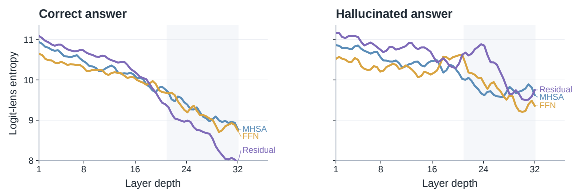

Figure 1: **TriLens summarizes pathway-wise logit-lens concentration across depth.** Supported answers show coordinated entropy sharpening across MHSA, FFN, and residual-stream readouts, whereas hallucinated answers tend to retain higher, less stable, and less synchronized entropy across depth.

1 Introduction

Large language models (LLMs) have reshaped a wide range of NLP tasks, but their tendency to generate hallucinations—outputs that are fluent yet factually wrong or inconsistent with provided context—remains a fundamental obstacle to deployment in knowledge-sensitive settings (Ji et al., 2023; Li et al., 2023). Reliable *detection* enables selective abstention, confidence calibration, and targeted retrieval.

Among existing detection approaches, *white-box* methods that analyze an LLM's internal state during a single forward pass are particularly attractive: they avoid the need for multiple samples or external references while enabling efficient inference-time scoring. Recent work has shown that hallucination-relevant information can be extracted from hidden states, gradients, semantic-entropy probes, spectral summaries, attention kernels, and update-consistency features (Azaria and Mitchell, 2023; Kossen et al., 2024; Hu et al., 2024; Chen et al., 2024; Sriramanan et al., 2024; Zhang et al., 2025; Yu et al., 2026b,a). Relatedly, hidden states can encode an evidence-supported causal direction even when the model's Yes/No output fails to express it (Ding and Zhang, 2026). However, these approaches still largely treat internal computation as a high-dimensional representation to be classified, rather than asking which low-dimensional aspect of the computation changes when a model moves toward an unsupported answer.

This leads to two empirical questions. First, can hallucination be detected from how logit-lens concentration changes across depth, rather than from final-layer confidence or high-dimensional hidden states? Second, do the main transformer pathways provide redundant views of that uncertainty, or do attention, feed-forward, and residual-stream readouts carry complementary evidence? A useful detector should answer these questions under a single

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forward pass, expose where uncertainty enters the computation, and avoid turning every hidden state into a large learned representation. We therefore test a simpler alternative: instead of storing activations themselves, we ask how uncertain different parts of the computation look when read through the model's own vocabulary lens.

TriLens records one scalar from each of three internal locations at every layer: the multi-head self-attention output, the feed-forward output, and the residual stream. Each location is projected through the logit lens, converted to a Shannon entropy, and concatenated across depth. The result is a compact $3L$-dimensional trajectory that tracks how logit-lens concentration changes across pathways. It is small enough for a simple probe, but structured enough to distinguish whether contextual routing, parametric recall, and the composed residual state sharpen together or remain diffuse.

Why should a first-order statistic suffice? Our working hypothesis is that hallucination-relevant errors are often accompanied by elevated uncertainty across internal pathways of an LLM, and that per-layer Shannon entropy of logit-lens distributions is a compact diagnostic correlate of this effect. In a decoder transformer, MHSA routes contextual information while the FFN pathway retrieves parametric content (Geva et al., 2021, 2023; Elhage et al., 2021). When these pathways align, their logit-lens distributions typically sharpen with depth; when they diverge, refinement can become slower and higher-entropy. TriLens tracks this behavior with three scalars per layer, yielding a compact alternative to high-dimensional hidden-state features.

Contributions. Our main contribution is to test these questions through a compact pathway-wise entropy feature, supported by ablations and diagnostic comparisons.

- We introduce **TriLens**, a per-layer logit-lens entropy feature over MHSA, FFN, and residual-stream states.
- Across three instruction-tuned LLMs and four English QA benchmarks, this $3L$-dimensional feature improves over the strong prior white-box baseline ICR Probe on *all 12* cells, averaging $+12.1$ AUROC, and attaining the highest AUROC in 11 of the 12 cells.
- Ablations show that MHSA, FFN, and residual-stream entropies contribute complementary signal, while an intra-layer module-disagreement term adds little.
- A comparison with DoLa-style cross-layer contrast features for detection shows that per-layer entropy captures much of the useful signal in that feature family.
- Layer-wise analysis shows that the peak-discriminative depth varies systematically across architectures and benchmarks, consistent with differences in how models integrate contextual and parametric information.

2 Related Work

White-box hallucination detection. The dominant white-box line trains detectors on internal activations from a single forward pass. SAPLMA (Azaria and Mitchell, 2023), layer-combination probes (CH-Wang et al., 2024), SEP (Kossen et al., 2024), gradient-aware detectors (Hu et al., 2024), and INSIDE (Chen et al., 2024) all fit this template. Training-free LLM-Check (Sriramanan et al., 2024) instead analyzes same-pass internal signals, while ICR Probe (Zhang et al., 2025) reframes detection around residual-stream updates. Entropy-Lens (Ali et al., 2025) analyzes layerwise entropy trajectories, and ReDeEP (Sun et al., 2025) studies internal mechanisms for RAG hallucination detection. TriLens differs by using a compact $3L$-scalar feature built from pathway-specific logit-lens entropy rather than high-dimensional hidden states or update-consistency scores.

Uncertainty-based detection. A parallel black-box line scores hallucination risk from a model's own uncertainty, including calibration-style signals (Kadavath et al., 2022; Malinin and Gales, 2021), semantic uncertainty and semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024), and recent refinements based on Bayesian estimation, pairwise semantic similarity, sampling- or fact-level self-consistency, or log-probability time series (Ciosek et al., 2025; Nguyen et al., 2025; Manakul et al., 2023; Sawczyn et al., 2026; Shapiro et al., 2026). TriLens instead uses a single white-box pass over internal activations, without multiple sampled generations or output-layer uncertainty alone (Xue et al., 2026a).

Logit-lens and mechanistic motivation. The logit lens and Tuned Lens project intermediate

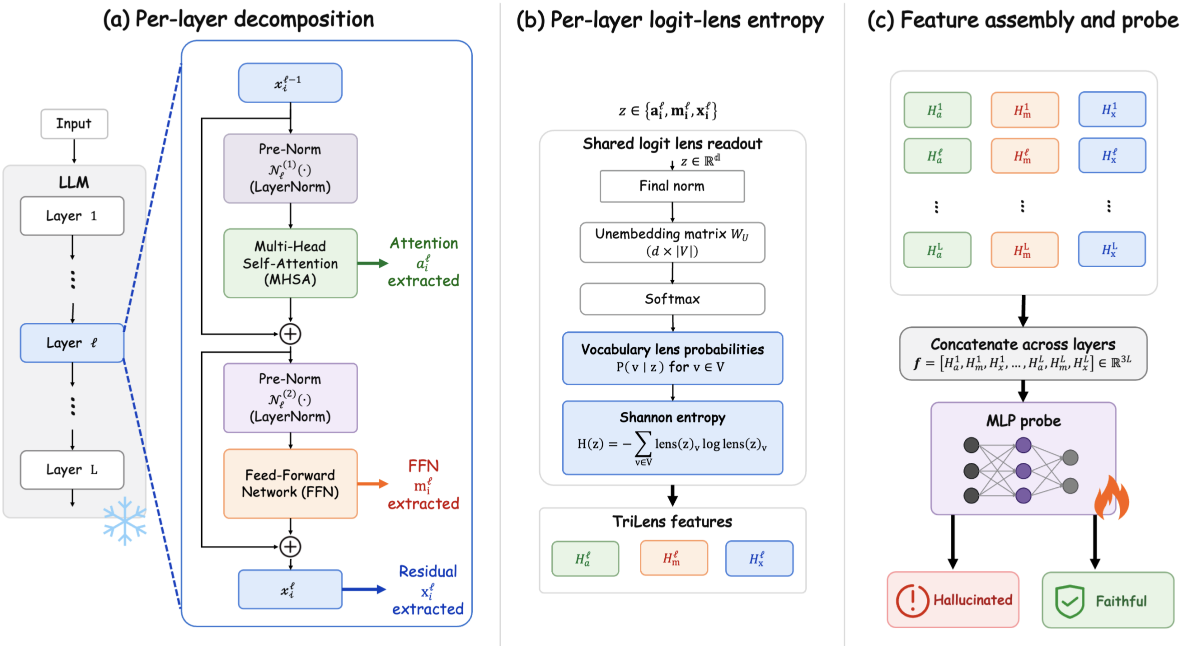

Figure 2: TriLens overview. A single forward pass over the evaluation sequence extracts the MHSA output, FFN output, and residual stream at every layer; each is projected through the model's logit lens, converted to a Shannon entropy, concatenated into a $3L$-dimensional feature, and fed to a probe.

states into vocabulary space (nostalgebraist, 2020; Belrose et al., 2023); DoLa (Chuang et al., 2024) contrasts these distributions across layers. Our design is further motivated by mechanistic views of FFNs as key-value memory, MHSA as residual-stream routing, and depth-varying module influence (Geva et al., 2021, 2023; Elhage et al., 2021; Stolfo et al., 2023). TriLens combines these ingredients into a supervised detection feature based on pathway-specific entropy trajectories; §4.6 compares it with DoLa-style cross-layer contrast features for detection.

3 Method

3.1 Per-Module Logit-Lens Entropy

We consider an LLM with L decoder layers and vocabulary V. At layer ℓ and token position i, the residual stream updates as

$$x_i^\ell = x_i^{\ell-1} + \underbrace{\mathrm{MHSA}^\ell(\mathcal{N}_\ell(x_i^{\ell-1}))}_{a_i^\ell} + \underbrace{\mathrm{FFN}^\ell(\mathcal{N}_\ell(x_i^{\ell-1} + a_i^\ell))}_{m_i^\ell}. \tag{1}$$

Here a_i^ℓ is the complete self-attention module output after the attention heads are combined and projected through the module's output projection W_O. It is therefore the vector actually written into the residual stream, not an unprojected concatenation of per-head outputs. For a hidden activation $z \in \{x_i^\ell, a_i^\ell, m_i^\ell\}$, the *logit lens* (nostalgebraist, 2020) projects z into vocabulary-space probabilities

$$\mathrm{lens}(z) = \mathrm{softmax}\big(W_U \, \mathrm{Norm}_{\mathrm{final}}(z)\big), \quad H(z) = -\sum_{v \in V} \mathrm{lens}(z)_v \log \mathrm{lens}(z)_v. \tag{2}$$

using the model's own unembedding matrix W_U and final normalization layer. Here $\mathcal{N}_\ell$ denotes the layer's pre-submodule normalization (e.g., RMSNorm in the model families studied here), while the lens itself always uses the model's *final* normalization layer before unembedding. This mirrors the standard logit-lens construction on the residual stream and keeps a^ℓ, m^ℓ, and x^ℓ in the same readout coordinate. In implementation we apply the final normalization directly to each of a^ℓ, m^ℓ, and x^ℓ, with no additional centering or rescaling. The *per-layer logit-lens entropy* is the Shannon entropy of this distribution; we compute it at all three positions, yielding $H_a^\ell, H_m^\ell, H_x^\ell$ for every layer (Eq. 2). TriLens applies the same readout not only to the residual stream but also to the isolated MHSA and FFN writes, using the resulting three-pathway trajectory as a detection feature.

3.2 Mechanistic Motivation

The choice to measure entropy *per module* rather than only on x^ℓ, and to use *Shannon entropy* rather than richer summary statistics, is motivated by three established views of transformer dynamics.

Dual-pathway decomposition. Under the residual-stream framework of Elhage et al. (2021), a^ℓ and m^ℓ are functionally distinct pathways: the OV circuit of MHSA copies content from earlier positions, while FFN functions as a read-out from a parametric key-value memory (Geva et al., 2021, 2023). In factual completion, both pathways often converge on the same target token—MHSA because the context uniquely determines the answer, FFN because the parametric memory has stored the relevant key–value pair. In hallucinated completion under our benchmark setting, the pathways can *diverge*: a^ℓ may still route contextually correct information, while m^ℓ can partially recall the same correct alternative even though the observed token is wrong. Projecting a^ℓ and m^ℓ independently through the logit lens exposes this divergence, which a single measurement on the composed x^ℓ can mask whenever the two pathways cancel in direction but not in magnitude. Although a^ℓ and m^ℓ are not themselves full residual states, they are additive writes in the same residual space and are therefore legitimate objects for the model's own final readout. The resulting distributions should be interpreted as *readout preferences of an isolated write* rather than as standalone next-token predictions; our claim is empirical and diagnostic, not that a^ℓ or m^ℓ alone constitute a full decoder state. Accordingly, the role of H_a^ℓ and H_m^ℓ in TriLens is not to replace H_x^ℓ but to supply complementary pathway-specific readouts under a shared lens.

Superposition and commitment. The residual stream at any layer is a linear superposition of all prior module writes. When the writes are coherent on a single vocabulary direction, the lens distribution is sharply peaked; when they are incoherent, the distribution spreads over multiple candidates. Shannon entropy is a compact statistic that tracks this spreading: it is invariant to which candidate tokens are involved, makes no assumption about whether the vocabulary is flat or structured, and is a calibrated function of the distribution's concentration. Richer alternatives such as Jensen–Shannon divergence to the final layer (the DoLa-style signal used in our detection control) or a spectral summary lose information about within-layer commitment, as we show empirically in §4.6.

Iterative refinement trajectories. The logit-lens view treats each layer as producing a refinement of a running next-token prediction (nostalgebraist, 2020; Belrose et al., 2023). Under this view, $H(\mathrm{lens}(z^\ell))$ is the uncertainty of the model's belief at depth ℓ along that module's pathway. For a correct continuation, uncertainty often decreases with depth as module contributions reinforce each other. For a hallucinated continuation, the refinement trajectory is more often non-monotone: uncertainty can re-enter the distribution at the depth where the parametric pathway recalls the true token while the forced context fixes the output on the wrong one. The per-layer vector $(H_a^\ell, H_m^\ell, H_x^\ell)_\ell$ can thus be viewed as a trajectory of this three-pathway refinement process, and a linear classifier on this trajectory can separate the two regimes in our benchmark setting.

These three views explain why the feature is minimal (three location-specific scalars per layer), why it is non-redundant (each scalar measures a different pathway's commitment), and why it is architecturally natural (it uses only the model's own unembedding and pre-trained norm). Section 4.3 examines the resulting empirical layer-wise trajectories.

3.3 Feature Construction

Given an evaluation sequence, we run the LLM once and extract $(H_a^\ell, H_m^\ell, H_x^\ell)$ at the token positions used for scoring. During extraction, this produces an $[L, T]$ trajectory for each readout over the T response positions. Unless otherwise stated, all main results use the first response token (i=0) as the fixed sample-level readout. This length-neutral rule avoids introducing response-length differences between correct and hallucinated samples. The resulting feature is

$$\mathbf{f} = (H_a^1, H_m^1, H_x^1, \dots, H_a^L, H_m^L, H_x^L) \in \mathbb{R}^{3L}. \tag{3}$$

For the models we consider, $3L$ ranges from 84 (Qwen2.5-7B, L=28) to 126 (Gemma-2-9B, L=42)—roughly 30× smaller than the raw hidden-state features used by SAPLMA and its descendants.

3.4 Probes

We evaluate two classifiers on $\mathbf{f}$: (i) a **linear probe** (L2-regularized logistic regression), which highlights that our feature is already separable without a learned transformation; and (ii) an **MLP probe** ($3L \to 128 \to 64 \to 32 \to 1$; LeakyReLU(0.01), BatchNorm, Dropout 0.3), using the same hidden-layer topology as the architecture used by the strong prior baseline (Zhang et al., 2025). Because TriLens uses a $3L$-dimensional input whereas ICR Probe uses an L-dimensional input, this comparison controls probe family and training recipe but does not fully equalize input dimensionality. We therefore treat the MLP comparison as a matched probe-family comparison rather than as a dimension-controlled capacity study, and we also report linear-probe results plus feature-subset ablations to separate feature quality from probe size. As an additional dimension-matched control, Appendix B compares TriLens against a repeated-feature baseline $H_x^{\times 3}$ that simply copies the residual-stream entropy three times to match the same $3L$ input size without adding new information, and further reports stricter MLP controls that reduce the three-pathway feature back to L dimensions via simple summaries or train-split PCA. Training details (BCE loss, Adam, 50 epochs, LR scheduler) are identical to those in Zhang et al. (2025) and are listed in Appendix H.

4 Experiments

4.1 Experimental Setup

Models. We evaluate on three recent instruction-tuned LLMs spanning the ∼7–9B parameter range and three architecture families: Qwen2.5-7B-Instruct (Qwen et al., 2025), Meta-Llama-3-8B-Instruct (Grattafiori et al., 2024), and Gemma-2-9B-it (Team et al., 2024). The first two use the standard RMSNorm+GQA decoder stack; Gemma-2 alternates full and sliding-window attention.

Datasets. We evaluate on four benchmarks commonly used in prior white-box detection work: HaluEval-QA (Li et al., 2023), SQuAD2.0 (Rajpurkar et al., 2018), HotpotQA (Yang et al., 2018), and TriviaQA (Joshi et al., 2017). For datasets that natively provide positive/negative supervision (HaluEval, SQuAD2), we use the provided supervision directly. For HotpotQA and TriviaQA, we follow a fixed benchmark-specific preprocessing pipeline to instantiate the evaluation examples used in our experiments. The same preprocessed instances are shared by TriLens and all baselines.

Metrics and protocol. Each benchmark is split 80/20 into train and test sets, and we report test-set AUROC averaged over five random seeds. Unless otherwise stated, we randomly sample 10,000 instances from each dataset, keeping the final label distribution balanced within every dataset. All model states are extracted from a single forward pass over the evaluation sequence, and the same sampled instances, splits, model suite, and model–dataset grid are shared by TriLens and all baselines. Further implementation details are listed in Appendix H.

Baselines. We compare against six baselines spanning both training-free and trainable white-box detectors. The training-free baselines are perplexity (**PPL**) (Ren et al., 2023), length-normalized entropy (**LN-Entropy**) (Malinin and Gales, 2021), and **LLM-Check** (Sriramanan et al., 2024); the trainable baselines are **SAPLMA** (Azaria and Mitchell, 2023), **SEP** (Kossen et al., 2024), and the previous state-of-the-art **ICR Probe** (Zhang et al., 2025). Appendix A.4 summarizes these baselines and provides a taxonomy that distinguishes supervision, model access, inference budget, and relative cost. All baseline results are produced under the same evaluation protocol, group-preserving 80/20 split, model suite, and model–dataset grid as TriLens.

4.2 Main Results

Table 1 reports test AUROC on all (model, dataset) cells against the six baselines. Our primary method, **TriLens (MLP)** (last row per model), uses the feature $\mathbf{f} = (H_a^\ell, H_m^\ell, H_x^\ell)_\ell$ with the same MLP probe topology used by ICR Probe. Unless otherwise noted, *TriLens* refers to this MLP variant used in Table 1.

Main finding. Hallucination risk is visible in layer-wise changes in logit-lens concentration: a compact trajectory of entropies is sufficient to separate supported from hallucinated answers in a single forward pass. TriLens improves over ICR Probe on all 12 cells and is the highest-AUROC entry in 11 of the 12 cells, with gains ranging from +4.2 AUROC (Qwen2.5-7B on TriviaQA) to +16.0 AUROC (Qwen2.5-7B on SQuAD2). The improvement is consistent across model families (+10.2 averaged over Gemma-2, +11.0 over Qwen2.5,

LLM	Methods	HaluEval	SQuAD2	HotpotQA	TriviaQA
Gemma-2-9B	PPL	0.5538	0.5348	0.7239	0.7536
	LN-Entropy	0.7357	0.6818	0.7349	0.7023
	LLM-Check	0.5780	0.5517	0.5102	0.5911
	SAPLMA	0.8123	0.7409	0.8329	0.7766
	SEP	0.6439	0.6627	0.6149	0.7726
	ICR Probe	<u>0.8436</u>	<u>0.8142</u>	<u>0.8409</u>	<u>0.8001</u>
	TriLens (Ours)	**0.9277**	**0.9270**	**0.9433**	**0.9106**
Qwen2.5-7B	PPL	0.5512	0.5278	0.6069	0.7041
	LN-Entropy	0.7286	0.6564	0.6913	0.6938
	LLM-Check	0.5367	0.5639	0.5518	0.5604
	SAPLMA	0.7725	0.7016	0.7689	**0.8297**
	SEP	0.6634	0.6418	0.6536	0.7449
	ICR Probe	<u>0.8076</u>	<u>0.7382</u>	<u>0.7865</u>	0.7751
	TriLens (Ours)	**0.9053**	**0.9061**	**0.9242**	<u>0.8103</u>
Llama-3-8B	PPL	0.5867	0.6472	0.6658	0.7085
	LN-Entropy	0.6574	0.6249	0.6661	0.5928
	LLM-Check	0.5263	0.5356	0.5559	0.5482
	SAPLMA	0.7315	0.7043	<u>0.7768</u>	0.7586
	SEP	0.7309	<u>0.7274</u>	0.6607	0.7048
	ICR Probe	<u>0.7671</u>	0.7568	0.7909	0.7396
	TriLens (Ours)	**0.9136**	**0.9169**	**0.9425**	**0.8861**

Table 1: **Main results**: test AUROC on four benchmark datasets under a matched evaluation protocol. **Bold** marks the best method in each model–dataset cell; <u>underlining</u> marks the second-best. TriLens improves over ICR Probe on all 12 cells and is best in 11, with average gain $+12.1$ AUROC. Per-cell standard deviations are listed in Appendix D.

and $+15.1$ over Llama-3) and across benchmark regimes ($+12.8$ over the three context-grounded tasks and $+10.2$ on TriviaQA).

Probe and capacity controls. The pattern is not explained solely by the MLP probe. A linear probe on the same features also outperforms ICR Probe on all 12 cells, with the MLP adding another ~ 3.2 AUROC points (Appendix B). Dimension-matched controls further show that simply repeating H_x three times is nearly identical to H_x alone, whereas TriLens remains better on all 12 linear-probe cells. A stricter MLP sweep with L-dimensional reductions of the three-pathway feature gives the same conclusion, indicating that the gain is not explained by input dimensionality alone.

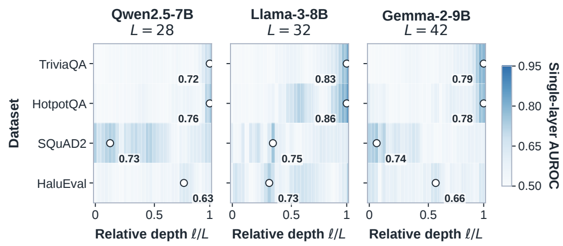

Figure 3: **Peak-discriminative depth varies across models and benchmarks.** Each cell reports $\max(\mathrm{AUROC}, 1-\mathrm{AUROC})$ obtained by using the first-response-token scalar H_x^ℓ directly as a score, with no trained single-layer probe. Columns denote relative depth ℓ/L; the peak cell in each row is circled. This analysis is descriptive and does not select or reweight layers in TriLens.

4.3 Per-Layer Analysis

Detailed trajectory plots and peak-layer densities are deferred to Appendix F. Figure 3 summarizes the descriptive single-layer analysis used to locate discriminative depth.

The peak-discriminative depth varies substantially across model–dataset pairs: some open-book cells peak in early or middle layers, while TriviaQA and several HotpotQA cells peak only near the final layer. This benchmark-conditional variation is consistent with prior mechanistic analyses (Geva et al., 2023; Zhang et al., 2025; Yu et al., 2026c), which suggest architecture- and task-dependent internal dynamics across depth.

4.4 Feature Ablation

The ablation answers whether the three readouts provide redundant or complementary evidence. We train the MLP probe on progressively larger subsets of $\mathbf{f}$ and additionally include an intra-layer module-disagreement term $\mathrm{JSD}_{am}^\ell = \mathrm{JSD}(\mathrm{lens}(a_i^\ell), \mathrm{lens}(m_i^\ell))$ as an additional fourth per-layer statistic, motivated by an intra-layer analog of ICR Probe's consistency construction. Detailed ablation numbers are reported in Appendix E. Figure 4 visualizes both the layer-wise separation and the feature-subset results.

The useful signal is not confined to the residual stream: adding either H_a or H_m to H_x improves performance on all 12 cells, and the full 3-entropy feature improves further in every cell. This indicates complementary predictive evidence across the three locations. Adding the intra-layer disagreement term changes AUROC by at most $+0.006$ per cell and $+0.001$ on average, suggesting little value beyond the entropy feature itself. The relative importance of H_a and H_m is architecture-dependent: the H_a+H_x branch is stronger on Gemma-2, whereas H_m+H_x is stronger on Llama-3 and part of Qwen2.5.

4.5 Cross-Dataset Diagnostics

The main results in Table 1 use a separate probe per benchmark. We next use two benchmark-shift diagnostics; detailed results appear in Appendix C. First, we train one probe per model on the union of all four benchmark training splits and evaluate it on each benchmark's held-out test split. This shared-probe setting still exceeds per-dataset ICR Probe on all 12 cells, with average gain $+10.7$ AUROC, while trailing our own per-dataset probes by only ~ 1.4 AUROC on average. Second, we train on one benchmark and test on another to form benchmark-level transfer heatmaps. In this cross-dataset transfer diagnostic, TriLens is the strongest method on all 12 off-diagonal cells and on all 16 cells overall; its off-diagonal mean AUROC is 0.848, compared with 0.738 for ICR Probe and 0.678 for SAPLMA. We therefore view the transfer result as further evidence that the gain is not confined to in-domain fitting, while still treating it as a distribution-shift diagnostic rather than as evidence of universal cross-benchmark generalization.

4.6 Comparison with Cross-Layer Contrast Signals

TriLens and DoLa (Chuang et al., 2024) both derive signals from logit-lens distributions but differ in what they measure: DoLa contrasts logit-lens distributions *across* layers and uses the result to modify decoding, while we measure entropy *within* each layer and use it for detection. To assess whether our per-layer entropy feature could be explained as a reformulation of the cross-layer contrast signal, we repurpose a DoLa-inspired cross-layer JSD quantity as a supervised detection feature. This experiment does not reproduce or evaluate DoLa's contrastive decoding procedure; it compares *DoLa-style cross-layer contrast features for detection* under the same probe protocol as TriLens.

Dataset	Full JSD	TriLens	TriLens+JSD
HaluEval	0.7623	0.9136	**0.9155**
SQuAD2	0.8538	0.9158	**0.9172**
HotpotQA	0.8928	**0.9425**	0.9422
TriviaQA	0.8679	**0.8861**	0.8860

Table 2: Comparison with full-vector DoLa-style cross-layer contrast features for detection (MLP probe, test AUROC). Full results appear in Appendix G.

Concretely, define the full-vector DoLa-style detection feature as

$$f^{\ell}_{\mathrm{DoLa}} = \mathrm{JSD}\big(\mathrm{lens}(x^{\ell}), \mathrm{lens}(x^{L})\big), \quad \ell = 0, \ldots, L-1. \tag{4}$$

and assemble the L-dimensional vector $\mathbf{f}_{\mathrm{DoLa}}$. We train the same MLP probe on (a) $\mathbf{f}_{\mathrm{DoLa}}$, (b) the TriLens feature $\mathbf{f}_{\text{3-ent}}$, and (c) their concatenation $\mathbf{f}_{\text{3-ent}} \oplus \mathbf{f}_{\mathrm{DoLa}}$. Experiments are run under the same group-preserving 80/20 evaluation protocol used in our main experiments, using the same 10k-scale feature files as the main paper. Table 2 reports the main-setting comparison, while Appendix G gives the full configuration grid.

Findings. Table 2 presents three observations. First, **our per-layer entropy consistently outperforms the DoLa-style cross-layer contrast feature on every dataset**, with gaps ranging from $+0.02$ to $+0.15$ AUROC and an average gain of roughly $+0.07$ in the main setting. Second, **their union changes performance only slightly and without a consistent upward pattern** relative to our feature alone: averaged across the four datasets, the union changes AUROC by less than $+0.001$. The cross-layer contrast signal therefore contributes little beyond what is already captured by per-layer entropy. Third, the DoLa-style feature degrades most severely on HaluEval, where in our default setup, intermediate layers have already converged onto the final layer's prediction (so the JSD collapses toward zero regardless of whether the response is hallucinated or correct); per-layer entropy is not subject to this degeneracy because it measures the shape of each layer's distribution independently. As a selection-rule control

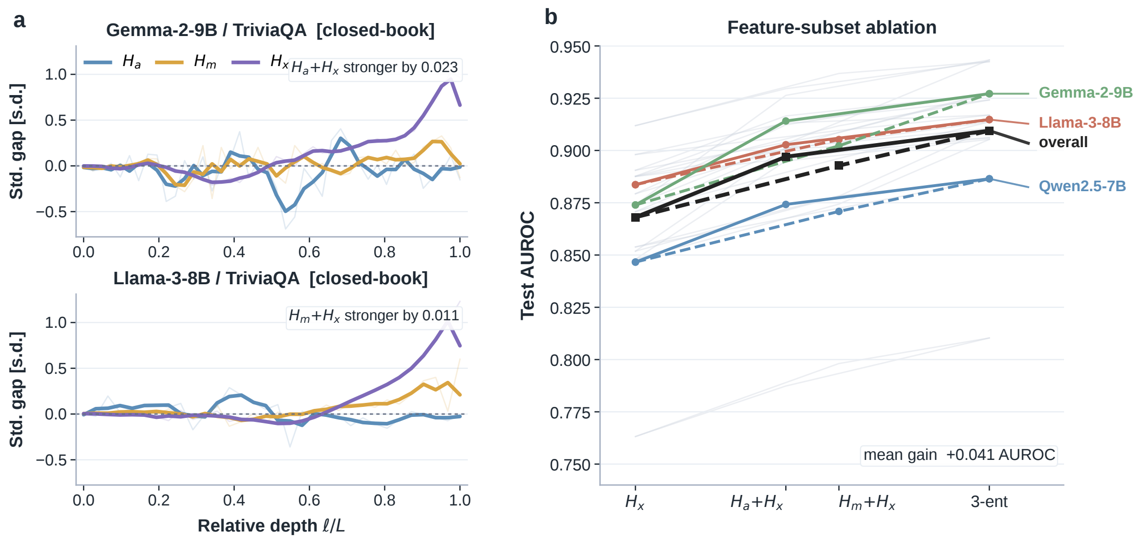

Figure 4: **The three entropy trajectories provide complementary predictive evidence. Left:** standardized per-layer separation between hallucinated and correct samples for H_a^ℓ, H_m^ℓ, and H_x^ℓ on two representative TriviaQA cells. The stronger two-feature branch is architecture-dependent. **Right:** feature-subset ablation across all 12 (model, dataset) cells. Colored lines show per-model means; grey lines show individual cells. Both two-feature branches improve over H_x alone, and the full three-entropy feature attains the best mean.

on Llama-3-8B across the four datasets, the full JSD vector averages 0.844 AUROC, whereas max-JSD and a train-selected single layer both average 0.786. The selected layer is chosen using only the training split. Appendix G reports these controls separately from DoLa decoding.

Additional scope and efficiency checks. Appendix I reports targeted Qwen3.5 checks across 0.8B–9B parameters and two non-QA HaluEval formats. TriLens reaches 0.910–0.926 AUROC on HaluEval-QA and 0.679/0.727 on summarization/dialogue, respectively, outperforming final-layer entropy in each setting. These are limited scope checks rather than a replacement for the protocol-matched main comparison. Appendix H.2 reports that layer-wise extraction is 14.91$\times$ slower than a standard forward pass in the measured setup, while increasing peak memory by only 0.18 GB.

5 Conclusion

We have presented *TriLens*, a $3L$-dimensional feature that computes Shannon entropies of logit-lens distributions at the MHSA output, FFN output, and residual stream at every layer. Across three mid-sized LLMs and four English QA benchmarks, under the same MLP probe family used by a strong prior baseline, this feature improves over ICR Probe on all 12 cells with an average gain of +12.1 AUROC and attains the highest AUROC in 11 of the 12 cells. Ablations have identified all three module entropies as independently informative and found little additional value from intra-layer module-disagreement. A comparison with DoLa-style cross-layer contrast features for detection suggests that TriLens captures much of the useful signal in this feature family. Layer-wise analysis further indicates that the most informative depth varies across architectures and benchmarks rather than concentrating at a universal layer. Targeted Qwen3.5 checks extend the observation across 0.8B–9B scales and two non-QA HaluEval formats, although the lower non-QA AUROCs reinforce the need for narrow task claims. These results establish TriLens as a mechanistically motivated diagnostic for the controlled QA-style settings evaluated here, not as causal evidence of how hallucinations arise in all generation tasks.

Limitations

TriLens provides a white-box risk score under the evaluated protocol, but it is not a factual verification system. False positives can reject correct answers, while false negatives can leave unsupported answers unflagged; these errors require particular caution in high-stakes applications. The method also requires internal activations and hallucination labels for probe training, restricting it to transparent models and supervised settings. Our main proto-

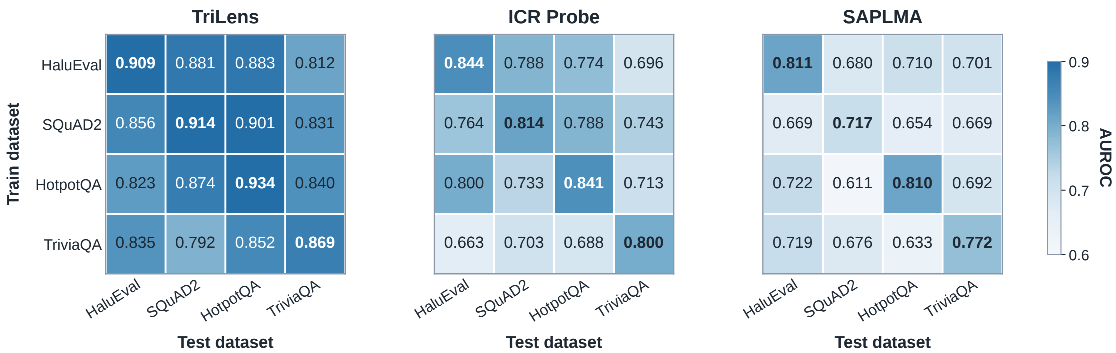

Figure 5: Cross-dataset generalization heatmaps for TriLens, ICR Probe, and SAPLMA. Each cell reports AUROC for the train-on-row, test-on-column setting. TriLens is strongest on all 12 off-diagonal cells and on all 16 cells overall, indicating more stable transfer under benchmark shift.

col covers English QA-style benchmarks and three mid-sized instruction-tuned models. Additional Qwen3.5 checks span 0.8B–9B and two non-QA HaluEval formats, but remain single-benchmark scope checks against final-layer entropy rather than broad cross-task validation. The cross-dataset experiments test benchmark shift, not universal deployment transfer. Long-form generation, RAG, multilingual and multimodal settings (Kang et al., 2026), and broader task coverage remain outside the validated scope. Finally, our study focuses on detection rather than mitigation and does not determine how the score should be used to reduce hallucinations during generation. Because the evaluated models are already instruction-tuned, we do not isolate how post-training failures or conflicts between supervised and pre-training knowledge shape the entropy trajectories (Xue et al., 2026b).

Ethics Statement

Our study is conducted entirely on publicly available LLMs and established public benchmarks, introduces no new human-subject data collection, and requires no additional annotation. We do not release any new text dataset containing personally identifying information or offensive content; all reported results are aggregate metrics over benchmark splits. We use the public models and benchmarks only under their publicly stated access conditions and licenses or terms of use, and our released code contains no redistributed model weights or benchmark text. Our use of these artifacts is limited to research evaluation of hallucination detection and is consistent with their benchmark or model-evaluation purpose and access conditions. We acknowledge the standard dual-use consideration that white-box access to internal representations can expose attack surfaces that may be exploited to refine adversarial generations (Xing et al., 2025, 2026). However, as a detection-only capability without any generation or mitigation component, the net effect is defensive.

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A Experimental Details

A.1 Computing Infrastructure

Our experiments were conducted on a server equipped with 6 NVIDIA GeForce RTX 4090 GPUs (24 GB memory each), running PyTorch with CUDA acceleration. Across multiple rounds of feature extraction, ablation, and evaluation experiments, the total computational budget amounted to 100–200 GPU hours. Feature extraction requires one teacher-forced forward pass per scored sequence; probe training is lightweight relative to LLM-state extraction and is repeated across seeds and evaluation configurations reported in the main text and appendix.

A.2 Details about Datasets

For our experiments, we randomly sampled 10,000 instances from each dataset. Specifically, we used the QA subset of HaluEval and the 'rc.nocontext' subset of TriviaQA. These benchmarks are English-language QA-style evaluation sets covering factoid question answering, unanswerable questions, multi-hop reasoning, and hallucinated QA responses; we do not use demographic annotations or perform demographic subgroup analysis. Each dataset is split into 80%-20% for training and testing. We train and test on each dataset, reporting the corresponding results. This experimental setup for baseline methods matches ours exactly.

A.3 Sampling, Splits, and Supervision

All trainable methods use group-preserving 80/20 train/test splits, and the same split assignment is shared by TriLens and all baselines. All reported results are averaged over five random seeds.

A.4 Details about Baselines

Table 3 organizes the detector families in three explicit blocks according to supervision and model access. Within each block, separate rows make the inference budget and detector-side computation directly comparable. Sampling-based methods are contextual comparisons rather than matched baselines because they require multiple generations and different semantic processing. Representative sampling-based examples are SelfCheckGPT (Manakul et al., 2023) and semantic entropy (Farquhar et al., 2024).

We summarize the six evaluated baselines in Table 1. The first three are training-free sample-level scorers, whereas the latter three are trainable white-box detectors. All are evaluated in the same pipeline as TriLens, on the same model–dataset grid and group-preserving 80/20 split. Scalar baselines are used directly as sample-level scores; trainable baselines retain their original feature definition and are fit on the corresponding training split. For the 10k-scale evaluation sets used in the main paper, HaluEval and SQuAD2 follow the dataset sizes reported in the main text.

PPL. Perplexity of the scored sequence under the same forward-pass setup. Higher perplexity indicates lower model confidence and is used directly as a hallucination score.

LN-Entropy. Length-normalized predictive entropy of the scored sequence. This baseline measures output uncertainty while partially controlling for raw response length.

LLM-Check. A training-free attention-kernel baseline that scores responses from structural statistics of self-attention kernels rather than hidden states.

SAPLMA. A supervised hidden-state baseline that trains an MLP classifier on internal activations from selected layers.

SEP. A probe-based hidden-state baseline motivated by semantic entropy, using learned detectors over internal representations rather than black-box semantic clustering over sampled outputs.

ICR Probe. The strongest prior white-box baseline. It derives one scalar feature per layer from residual-stream update statistics and feeds the resulting layer-wise vector to the same 4-layer MLP architecture used throughout our comparisons; the shared probe hyperparameters are listed in Appendix H.

A.5 Probe Comparability Note

TriLens and ICR Probe are compared under the same training protocol and the same hidden-layer MLP topology, but not under equal input dimensionality: TriLens uses $3L$ input features whereas ICR Probe uses L. The main text therefore interprets the comparison as a feature-plus-probe comparison within a matched probe family, rather than as a fully dimension-controlled capacity study.

Method(s)	Training labels?	Internal access	Model calls per input	Detector computation	Relative cost
A. Supervised white-box probing — a learned detector is fitted on labeled internal features.					
TriLens	Yes	Module outputs, residual stream, and LM head	One scored sequence (one forward pass)	Trained MLP or linear probe	High (14.91× measured)
SAPLMA, SEP, ICR Probe	Yes	Selected hidden states or residual-update statistics	One scored sequence (one forward pass)	Trained probe	Medium
B. Training-free white-box scoring — a fixed score is computed from accessible model quantities.					
PPL, LN-Entropy	No	Token logits	One scored sequence (one forward pass)	Fixed scalar reduction	Low
LLM-Check	No	Self-attention kernels	One scored sequence (one forward pass)	Kernel statistics	Low–medium
C. Black-box sampling — uncertainty is estimated from multiple generated outputs without hidden-state access.					
SelfCheckGPT	No	Generated text only	$K > 1$ sampled responses	Cross-sample consistency	High
Semantic entropy	No	Samples and sequence likelihoods	$K > 1$ sampled responses	Semantic clustering and entropy	High

Table 3: Taxonomy of hallucination-detector families. "Training labels" indicates whether hallucination labels are required to fit the detector; K denotes the number of sampled responses. Only blocks A and B are evaluated under our matched single-sequence pipeline. Block C is included to clarify the additional generation and semantic-processing cost of sampling-based detection. Relative cost refers to inference time and excludes one-time probe training.

B Probe Controls and Fairness Checks

Table 4 reports the same TriLens feature used in the main paper, replacing the MLP probe with L2-regularized logistic regression under the same feature configuration as the main results. The qualitative conclusion is unchanged: the linear probe already improves over ICR Probe on all 12 cells, with the MLP adding a further $\sim$ 3.2 AUROC points on average. To isolate the effect of input dimensionality, Table 5 adds a dimension-matched control $H_x^{\times 3}$ that repeats the residual-stream entropy feature three times, yielding the same $3L$ input size as TriLens without introducing any new signal. Table 6 extends this check under the same MLP probe family with stricter dimensionality controls: three handcrafted L-dimensional reductions of the three-pathway feature and a train-split PCA projection of the full $3L$-dimensional TriLens vector back to L. We further report alternative readout controls in Appendix B.1, comparing the pathway-separated TriLens readout against intermediate-state and additive-write readouts built from the same layer-wise logit-lens pipeline.

B.1 Alternative Readout Controls

The main TriLens feature applies the final logit-lens readout to the isolated MHSA write a^ℓ, the isolated FFN write m^ℓ, and the composed residual state x^ℓ. To test whether the gains depend on this particular choice of readout location, we compare it with two more conservative alternatives built from the same layer-wise pipeline: (i) H_{pre}, the entropy of the intermediate state $x^{\ell-1} + a^\ell$ after the attention write but before the FFN write, and (ii) H_p, the entropy of the additive-write state $a^\ell + m^\ell$. We then pair each alternative with the standard residual-stream readout H_x and evaluate the resulting features under the same protocol as the main paper.

Three patterns are consistent across probe families. First, performance improves monotonically as the readout moves from residual-only H_x to $H_{\text{pre}} + H_x$, then to $H_p + H_x$, and finally to the full pathway-separated TriLens feature. Under the linear probe, the corresponding average AUROC increases from 0.814 to 0.833, 0.860, and 0.878, with TriLens strongest on all 12 cells. Second, the same ordering largely persists under the MLP probe, where TriLens is strongest on 11 of 12 cells and improves over H_x by $+0.052$ AUROC on average. Third, the mixed-state alternatives do help, but they remain consistently below the explicit three-way decomposition. We therefore interpret this sweep as evidence that TriLens benefits from separating MHSA, FFN, and residual-stream readouts,

LLM	Methods	HaluEval	SQuAD2	HotpotQA	TriviaQA
Gemma-2-9B	ICR Probe	0.8436	0.8142	0.8409	0.8001
	TriLens (Linear)	**0.8963**	**0.9004**	**0.9203**	**0.8864**
Qwen2.5-7B	ICR Probe	0.8003	0.7456	0.7917	0.7684
	TriLens (Linear)	**0.8597**	**0.8696**	**0.8746**	**0.7780**
Llama-3-8B	ICR Probe	0.7603	0.7634	0.7982	0.7325
	TriLens (Linear)	**0.8731**	**0.8806**	**0.9253**	**0.8700**

Table 4: Linear-probe results on the same 12 cells as Table 1. All TriLens entries are 5-seed means; seed standard deviation is ≤ 0.007 in every cell.

LLM	Dataset	H_x	$H_x^{\times 3}$	TriLens
Gemma-2-9B	HaluEval	0.7970	0.7970	**0.8963**
	SQuAD2	0.8497	0.8497	**0.9004**
	HotpotQA	0.7812	0.7813	**0.9203**
	TriviaQA	0.8238	0.8238	**0.8864**
Qwen2.5-7B	HaluEval	0.7746	0.7746	**0.8597**
	SQuAD2	0.8389	0.8389	**0.8696**
	HotpotQA	0.7992	0.7992	**0.8746**
	TriviaQA	0.7384	0.7384	**0.7780**
Llama-3-8B	HaluEval	0.8247	0.8248	**0.8731**
	SQuAD2	0.8254	0.8255	**0.8806**
	HotpotQA	0.8717	0.8717	**0.9253**
	TriviaQA	0.8385	0.8385	**0.8700**

Table 5: Dimension-matched fairness control under the linear probe. $H_x^{\times 3}$ repeats the L-dimensional residual-stream entropy feature three times to match TriLens's $3L$ input size. The repeated baseline is nearly identical to H_x, while TriLens remains better on all 12 cells.

rather than from using an arbitrary auxiliary lens location.

C Cross-Dataset Generalization Heatmaps

The main paper briefly reports the shared-probe variant of TriLens. Table 8 gives the corresponding per-benchmark numbers. We additionally report benchmark-level cross-dataset generalization heatmaps for TriLens, ICR Probe, and SAPLMA. In each 4×4 matrix, a probe is trained on the row benchmark and evaluated on the column benchmark under the same train/test transfer protocol.

Aggregate statistics. Across the 12 off-diagonal cells, TriLens reaches a mean AUROC of 0.848, compared with 0.738 for ICR Probe and 0.678 for SAPLMA. Its diagonal mean is also higher (0.907 vs. 0.825 and 0.778), so the advantage is not limited to either in-domain or out-of-domain evaluation alone.

Transfer pattern. The transfer gap is not perfectly uniform across source benchmarks. For TriLens, probes trained on SQuAD2 or HaluEval give the strongest off-diagonal averages (0.863 and 0.859), while TriviaQA is the weakest source row (0.826 off-diagonal mean). Even so, all 12 off-diagonal TriLens cells remain above 0.79, whereas the baselines show broader degradation. We therefore interpret Figure 5 as evidence of comparatively stable transfer under benchmark shift, while still viewing cross-dataset evaluation as a diagnostic rather than as a substitute for the in-domain results in Table 1.

D Main-Table Standard Deviations

Table 9 lists the standard deviations for the TriLens entries in Table 1, across the same five seeds used in the main-paper evaluation.

E Feature Ablation

Table 10 reports the full feature-set ablation from the main paper discussion.

Table 11 lists the standard deviations across the same five seeds used in Table 10.

F Per-Layer Diagnostics

This appendix collects the layer-wise plots and the fuller interpretation omitted from the compressed main text.

Peak-layer analysis is included only as a descriptive diagnostic rather than as a tuned component of TriLens. For each layer ℓ, we use the first-response-token scalar H_x^ℓ directly as the score and compute AUROC against the hallucination label; no single-layer probe is trained. Because the score direction can differ across settings, Figures 6 and 3 report $\max(\mathrm{AUROC}, 1 - \mathrm{AUROC})$. The main results always use the full per-layer feature vector, without selecting or reweighting layers.

LLM	Dataset	H_x	$H_x^{\times 3}$	TriMeanL	TriMaxL	TriMinL	TriLens+PCA→ L	TriLens
Gemma-2-9B	HaluEval	0.8325	0.8420	0.8364	0.8140	0.8188	0.8794	**0.8967**
	SQuAD2	0.8532	0.8638	0.8461	0.8319	0.8349	0.8773	**0.8951**
	HotpotQA	0.8051	0.8272	0.8326	0.7991	0.8050	0.8927	**0.9145**
	TriviaQA	0.8237	0.8313	0.8057	0.7681	0.8053	0.8714	**0.8838**
Qwen2.5-7B	HaluEval	0.8000	0.8131	0.8004	0.7534	0.7877	0.8379	**0.8576**
	SQuAD2	0.8523	0.8606	0.8358	0.8134	0.8202	0.8669	**0.8743**
	HotpotQA	0.8376	0.8529	0.8155	0.7813	0.7585	0.8655	**0.8789**
	TriviaQA	0.7330	0.7387	0.6933	0.6218	0.6934	0.7667	**0.7700**
Llama-3-8B	HaluEval	0.7635	0.7783	0.7874	0.7480	0.7840	0.8209	**0.8379**
	SQuAD2	0.8100	0.8383	0.7902	0.7469	0.8077	0.8093	**0.8448**
	HotpotQA	0.8612	0.8710	0.8737	0.8057	0.8539	0.9034	**0.9156**
	TriviaQA	0.8305	0.8326	0.8130	0.6753	0.8110	0.8471	**0.8578**

Table 6: Stricter dimensionality controls under the MLP probe. Besides $H_x^{\times 3}$, we compare three handcrafted L-dimensional reductions of the three-pathway feature and a train-split PCA projection of the full $3L$-dimensional TriLens vector back to L. Full TriLens remains strongest on all 12 cells.

Readout	Linear		MLP	
	Avg	Wins	Avg	Wins
H_x	0.8137	0/12	0.8169	0/12
$H_{\text{pre}} + H_x$	0.8334	0/12	0.8268	0/12
$H_p + H_x$	0.8604	0/12	0.8496	1/12
TriLens	**0.8779**	**12/12**	**0.8690**	**11/12**

Table 7: Readout-location robustness summary across the 12 model–dataset cells. "Avg" denotes mean test AUROC; "Wins" counts how often a readout is the strongest cell-wise configuration. All entries are 5-seed means under the same protocol as the main paper.

Model	Test	ICR[†]	Multi (ours)	Δ
Qwen2.5-7B	HaluEval	0.8003	0.8856	+0.085
	SQuAD2	0.7456	0.8915	+0.146
	HotpotQA	0.7917	0.9102	+0.118
	TriviaQA	0.7684	0.7998	+0.031
Llama-3-8B	HaluEval	0.7603	0.8965	+0.136
	SQuAD2	0.7634	0.9045	+0.141
	HotpotQA	0.7982	0.9329	+0.135
	TriviaQA	0.7325	0.8739	+0.141
Gemma-2-9B	HaluEval	0.8436	0.9132	+0.070
	SQuAD2	0.8142	0.9129	+0.099
	HotpotQA	0.8409	0.9225	+0.082
	TriviaQA	0.8001	0.8971	+0.097
avg		0.7883	**0.8950**	**+0.107**

Table 8: **Multi-dataset probe**: one probe per model, trained on the union of all four benchmarks and evaluated on each held-out test split. Each cell is a 5-seed mean (std $\leq$ 0.008). [†]ICR entries are our per-dataset reruns from Table 1.

Context-grounded vs. closed-book regimes. On context-grounded benchmarks such as SQuAD2 and HotpotQA, the correct–hallucinated gap often opens earlier and remains visible across much of the upper stack. On TriviaQA, by contrast, the trajectories tend to remain close until the final layers, consistent with a closed-book setting in which the relevant parametric recall signal is resolved late.

Architecture-dependent peak depth. Figure 3 shows that the peak-discriminative layer is not universal across architectures. Some Qwen2.5 cells peak in earlier or mid layers, whereas Llama-3 and Gemma-2 frequently peak closer to the final layers. This pattern is consistent with prior observations that different model families accumulate factual recall at different depths.

Pathway asymmetry. The reversal between Gemma-2 and Llama-3 in Figure 4 suggests that the relative usefulness of H_a and H_m depends on how readable each pathway's contribution is under the model's own unembedding. Gemma-2's hybrid attention stack may shift that balance toward MHSA relative to the other two models.

Smallest-gain cell. The smallest gain in the main table occurs on Qwen2.5-7B / TriviaQA. Its late-layer trajectories show weaker separation than the corresponding cells for Llama-3 and Gemma-2, suggesting a more diffuse parametric recall signal in this closed-book setting.

Table 12 reports the peak-discriminative layer for the direct scalar score based on H_x^{ℓ}, using the same setup as Figures 6–3. Layer indices are zero-based to match the figure annotations.

G DoLa-Style Detection Controls

Table 13 provides the full configuration grid behind Table 2: both probe families and all four datasets,

Model	Dataset	Std
Gemma-2-9B	HaluEval	0.0022
Gemma-2-9B	SQuAD2	0.0025
Gemma-2-9B	HotpotQA	0.0032
Gemma-2-9B	TriviaQA	0.0027
Qwen2.5-7B	HaluEval	0.0053
Qwen2.5-7B	SQuAD2	0.0025
Qwen2.5-7B	HotpotQA	0.0015
Qwen2.5-7B	TriviaQA	0.0051
Llama-3-8B	HaluEval	0.0014
Llama-3-8B	SQuAD2	0.0037
Llama-3-8B	HotpotQA	0.0033
Llama-3-8B	TriviaQA	0.0037

Table 9: Seed standard deviations for the TriLens entries in Table 1.

Model	Data	H_x	H_a+H_x	H_m+H_x	3-ent	3-ent+JSD$^{\ell}_{am}$
Gemma-2	HaluEval	.8793	.9123	.9087	.9277	**.9286**
	SQuAD2	.8905	.9165	.9085	.9270	**.9285**
	HotpotQA	.8741	.9264	.9135	.9433	**.9451**
	TriviaQA	.8518	.9013	.8782	.9106	**.9115**
Qwen2.5	HaluEval	.8480	.8950	.8743	.9053	**.9067**
	SQuAD2	.8876	.9006	.8992	.9061	**.9065**
	HotpotQA	.8877	.9138	.9118	.9242	**.9256**
	TriviaQA	.7632	.7874	.7981	**.8103**	.8102
Llama-3	HaluEval	.8708	.9008	.8962	.9136	**.9194**
	SQuAD2	.8981	.9132	.9095	.9169	**.9194**
	HotpotQA	.9119	.9296	.9368	.9425	**.9444**
	TriviaQA	.8538	.8675	.8783	**.8861**	.8846

Table 10: Feature-set ablation (MLP probe, test AUROC). **3-ent** $= (H_a, H_m, H_x)$; the last column adds JSD$^{\ell}_{am}$.

using the same supervised cross-layer contrast feature as the main paper. This is a detection-feature comparison, not an evaluation of DoLa decoding.

Model	Data	H_x	H_a+H_x	H_m+H_x	3-ent	3-ent+JSD$^{\ell}_{am}$
Gemma-2	HaluEval	0.004	0.002	0.003	0.002	0.002
	SQuAD2	0.006	0.004	0.006	0.004	0.004
	HotpotQA	0.003	0.003	0.003	0.004	0.004
	TriviaQA	0.004	0.004	0.004	0.004	0.005
Qwen2.5	HaluEval	0.005	0.002	0.005	0.003	0.003
	SQuAD2	0.006	0.005	0.004	0.006	0.005
	HotpotQA	0.005	0.005	0.003	0.004	0.005
	TriviaQA	0.006	0.004	0.008	0.006	0.006
Llama-3	HaluEval	0.003	0.003	0.001	0.002	0.002
	SQuAD2	0.003	0.004	0.004	0.004	0.004
	HotpotQA	0.004	0.003	0.004	0.002	0.003
	TriviaQA	0.003	0.003	0.005	0.004	0.004

Table 11: Standard deviations corresponding to Table 10 (MLP probe, same aggregation as the main results).

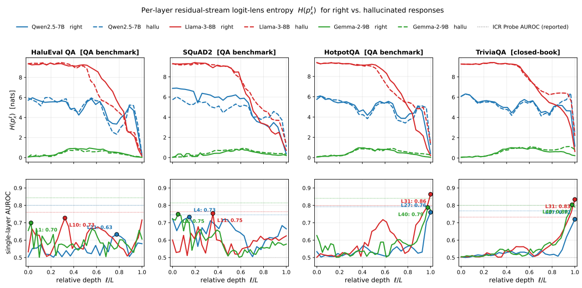

Figure 6: **Per-layer residual-stream logit-lens entropy** H_x^ℓ **(top)** and **descriptive single-layer AUROC (bottom)** across three models and four benchmarks. AUROC is computed directly from the first-response-token scalar, with no trained probe, and shown as $\max(\text{AUROC}, 1-\text{AUROC})$. Solid lines denote correct responses; dashed lines denote hallucinated responses.

Model	Dataset	ℓ^*	AUROC
Qwen2.5-7B	HaluEval	L21	0.6332
Qwen2.5-7B	SQuAD2	L4	0.7329
Qwen2.5-7B	HotpotQA	L27	0.7603
Qwen2.5-7B	TriviaQA	L27	0.7199
Llama-3-8B	HaluEval	L10	0.7274
Llama-3-8B	SQuAD2	L11	0.7541
Llama-3-8B	HotpotQA	L31	0.8638
Llama-3-8B	TriviaQA	L31	0.8340
Gemma-2-9B	HaluEval	L1	0.6994
Gemma-2-9B	SQuAD2	L2	0.7491
Gemma-2-9B	HotpotQA	L40	0.7881
Gemma-2-9B	TriviaQA	L40	0.8033

Table 12: Peak-discriminative layer ℓ^* and the corresponding descriptive $\max(\mathrm{AUROC}, 1-\mathrm{AUROC})$ for the first-response-token scalar H_x^ℓ, computed without a probe.

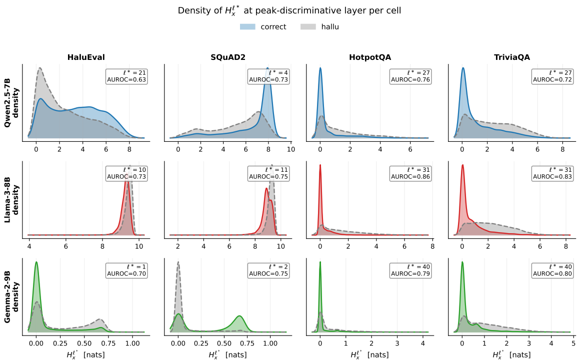

Figure 7: Probability density of $H_x^{\ell^*}$ at the peak-discriminative layer for correct vs. hallucinated responses.

Probe	Dataset	Full JSD	TriLens	TriLens+JSD
MLP	HaluEval	0.7623	0.9136	**0.9155**
	SQuAD2	0.8538	0.9158	**0.9172**
	HotpotQA	0.8928	**0.9425**	0.9422
	TriviaQA	0.8679	**0.8861**	0.8860
Linear	HaluEval	0.6790	0.8731	**0.8738**
	SQuAD2	0.7189	0.8778	**0.8798**
	HotpotQA	0.8111	0.9253	**0.9256**
	TriviaQA	0.8383	0.8700	**0.8710**

Table 13: Full grid for the DoLa-style cross-layer contrast feature for detection under both probe families.

Detection feature	Avg. AUROC	Layer choice
Full JSD vector	0.844	All layers
Max-JSD	0.786	Test-time max
Train-selected single layer	0.786	Training split only

Table 14: Selected-layer controls for DoLa-style cross-layer contrast features on Llama-3-8B across four datasets (MLP probe, first-response-token aggregation). The train-selected layer is chosen without test-set access.

H Efficiency and Complexity Analysis

This section reports the complete probe configuration and distinguishes compact stored features from the cost of layer-wise logit-lens extraction.

H.1 Training Hyperparameters

Table 15 lists the hyperparameters used for every main TriLens run; no configuration is selected on the test split.

H.2 Measured Runtime and Peak Memory

Table 16 reports wall-clock extraction cost on 100 Llama-3-8B-Instruct/HaluEval samples under the hardware and software setup described in Appendix A. The feature remains compact in memory, but the repeated vocabulary projections make extraction substantially slower than a standard forward pass.

H.3 Time Complexity

For a decoder with L layers and hidden size d, TriLens adds a logit-lens readout at three locations per layer and reduces each readout to a scalar entropy. Relative to methods that store or classify high-dimensional hidden states, the resulting sample representation is compact: TriLens produces a $3L$-dimensional vector, whereas hidden-state baselines typically operate on features that scale with d or selected layer subsets of size comparable to d. Probe-time complexity is therefore negligible compared with the upstream LLM forward pass; in practice, extraction dominates runtime and the downstream classifier is cheap to retrain across seeds.

H.4 Memory Footprint of Logit-Lens Extraction

For a model with vocabulary size $|V|$, a naive implementation could materialize, for each scored token, $3L$ intermediate logit-lens distributions of the form $\mathrm{softmax}(W_U \cdot \mathrm{Norm}(z)) \in \mathbb{R}^{|V|}$. In float32, this corresponds to a peak memory cost of $3L|V| \times 4$ bytes per token if all such distributions are retained simultaneously. For example, for Llama-3-8B (L=32, $|V|$≈128K), the raw peak is approximately $3 \times 32 \times 128000 \times 4 \approx 49$ MB per token. In our implementation, however, each distribution is used immediately to compute its Shannon entropy and is then discarded, so the persistent storage scales as $O(3L)$ rather than $O(3L|V|)$. For Llama-3-8B, this reduces the retained per-token state to only 96 scalar entropy values.

Model weights dtype	bfloat16
Logit-lens projection dtype	float32
Attention implementation	SDPA
Temperature τ	1.0
Max response tokens	32
Train/test split	80/20 group-preserving split
Seeds	5
Main-paper dataset sizes	reported in Section 4.1
Row labels per dataset	exactly balanced correct/hallucinated labels
Linear probe	L2 logreg, $C = 1$
MLP probe architecture	$3L \to 128 \to 64 \to 32 \to 1$
MLP activation	LeakyReLU(0.01)
MLP regularization	BatchNorm + Dropout 0.3
MLP optimizer	Adam, lr $= 5 \times 10^{-4}$
MLP training	50 epochs, batch 32
MLP LR schedule	ReduceLROnPlateau (p=5, f=0.5)

Table 15: Training hyperparameters used in the main experiments.

Setting	Latency/sample	Peak memory
Standard forward	36.82 ms	17.22 GB
TriLens extraction	549.10 ms	17.40 GB
Relative overhead	14.91×	+0.18 GB

Table 16: Measured extraction latency and peak memory over 100 Llama-3-8B-Instruct/HaluEval samples. TriLens has small additional peak memory but nontrivial layer-wise projection cost.

I Additional Model and Task Scope Checks

We use Qwen3.5 instruction-tuned checkpoints to test whether the feature remains useful outside the original model suite and QA-only task format. These experiments use the same first-response-token readout and supervised TriLens probe, but are reported as targeted scope checks rather than as protocol-matched extensions of Table 1.

Qwen3.5 setting	Final entropy	TriLens	Δ
0.8B / HaluEval-QA	0.566	0.910	+0.345
4B / HaluEval-QA	0.689	0.915	+0.225
9B / HaluEval-QA	0.648	0.926	+0.278
4B / HaluEval-Summarization	0.541	0.679	+0.138
4B / HaluEval-Dialogue	0.589	0.727	+0.138

Table 17: Targeted model-scale and non-QA scope checks (test AUROC). Δ is computed from the underlying unrounded runs. The non-QA results are positive but substantially weaker than the QA results and should not be read as broad long-form generalization.